

Nahyuk Lee

POSITION **Research Scientist**
VUNO Inc.

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RESEARCH INTERESTS **Computer Vision & Machine Learning**, in particular, the problems of 3D computer vision:

- Learning visual correspondences (*i.e.*, 2D & 3D feature matching) and its applications
- 3D point cloud registration & Geometric shape assembly
- Geometric deep learning (*i.e.*, equivariant learning)
- Representation learning for 3D generative model

EDUCATION **Pohang University of Science and Technology (POSTECH)**, Pohang, Korea
M.S. in Computer Science and Engineering **Feb 2023 - Feb 2025**

- Thesis: *Combinative Matching for Geometric Assembly*
- Advisor: [Prof. Minsu Cho](#)
- Admission with the highest distinction (**Bang Sung-Yang Fellowship**)

Chung-Ang University, Seoul, Korea
B.S. in Computer Science and Engineering **Mar 2019 - Feb 2023**

- Advisor: [Prof. Junseok Kwon](#)
- Cumulative GPA: 4.31/4.5 (Summa Cum Laude, **1st in Class of 2023**)

EXPERIENCE **Computer Vision and Geometry Group, ETH Zurich**, Zurich, Switzerland (Remote)
Research Collaborator **Oct 2025 - now**

- Advisor: [Dr. Sunghwan Hong](#) and [Prof. Dr. Marc Pollefeys](#)
- Researching 3D representation alignment and foundation-model-driven 3D perception for shape assembly and scene understanding [[ECCV 2026](#), [arXiv 2026](#)].

VUNO Inc., Seoul, Korea
Research Scientist (Alternative Military Service) **Jan 2025 - now**

- Extending prior 3D vision research (3D point cloud registration and equivariant learning) to medical image analysis.
- Led key components in VUNO Med[®]-DeepBrain[®] and VUNO Med[®]-LungCT AI[™].

Computer Vision Lab, POSTECH, Pohang, Korea
Graduate Research Assistant **Jul 2022 - Dec 2024**

- Advised by [Prof. Minsu Cho](#) and [Prof. Jaesik Park](#)
- Researched problems in 3D point cloud matching, with a particular focus on point cloud registration, geometric shape assembly, and equivariant learning [[ICCV 2025 Highlight](#), [ICML 2024](#), [IPIU 2024 Best Paper](#)].

PUBLICATIONS * denotes equal contribution. P: Preprint, C: Conference, W: Workshop, J: Journal

Selected publications (CV/ML/Robotics)

- [P1] Zhiang Chen*, **Nahyuk Lee***, Boyang Sun, Taein Kwon, Marc Pollefeys, Zuria Bauer, Sunghwan Hong, “PROSE: Training-Free Egocentric Scene Registration with Vision-Language Models,” *arXiv 2026*, *under review*
- [C5] **Nahyuk Lee***, Zhiang Chen*, Marc Pollefeys, Sunghwan Hong, “TORA: Topological Representation Alignment for 3D Shape Assembly,” *in Proceeding of the European Conference on Computer Vision (ECCV)*, 2026

- [C4] **Nahyuk Lee***, Juhong Min*, Junhong Lee, Chunghyun Park, Minsu Cho, “Combinative Matching for Geometric Shape Assembly,” in *Proceeding of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025, **Highlight (263/11, 152 $\approx$ 2.35%)**
- [C3] **Nahyuk Lee***, Juhong Min*, Junha Lee, Seungwook Kim, Kanghee Lee, Jaesik Park, Minsu Cho, “3D Geometric Shape Assembly via Efficient Point Cloud Matching,” in *Proceedings of the International Conference on Machine Learning (ICML)*, 2024
- [W1] Seung Hyun Lee, **Nahyuk Lee**, Chanyoung Kim, Wonjeong Ryoo, Jinkyu Kim, Sang Ho Yoon, Sangpil Kim, “Audio-Guided Image Manipulation for Artistic Paintings,” in *NeurIPS Workshop on Machine Learning for Creativity and Design (NeurIPS Workshop)*, 2021

Other publications (Interdisciplinary)

- [J3] **Nahyuk Lee**, Taemin Lee, “Bi-directional Point Flow Estimation with Multi-scale Attention for Deformable Lung CT Registration,” in *Applied Sciences*, 2025
- [C2] Doohyun Park, **Nahyuk Lee**, Sungjoo Lim, “Attention-based Interpretable Deep Learning with Radiomic Features for Pulmonary Nodule Classification,” in *Medical Imaging and Deep Learning (MIDL)*, Short Paper, 2025
- [J2] **Nahyuk Lee**, Taemin Lee, “The Study for Emotion Analysis from Images depend on Machine Learning Models,” in *Journal of Digital Contents Society (JDSCS)*, 2021
- [C1] Taemin Lee, **Nahyuk Lee**, Sanghyun Seo, Dongwann Kang, “A Study on the Prediction of Emotion from Image by Time-flow depend on Color Analysis,” in *Proceeding of the International Conference on Computational Science and Computational Intelligence (CSCI)*, 2020
- [J1] **Nahyuk Lee**, Kyungtaek Lee, Youngsup Park, Sanghyun Seo, Taemin Lee, “Development of 3D Reconstruction and Object Recognition Model using Video,” in *Journal of Digital Contents Society (JDSCS)*, 2020

PROFESSIONAL SERVICES

Reviewer of international journals

Transaction on Pattern Analysis and Machine Intelligence (**TPAMI**)

IF: 20.8

Reviewer of international conferences

Neural Information Processing Systems (**NeurIPS**)

International Conference on Learning Representations (**ICLR**)

European Conference on Computer Vision (**ECCV**)

International Conference on 3D Vision (**3DV**)

LSRW Workshop, Conference on Robot Learning (**CoRLW**)

WiCV Workshop, Asian Conference on Computer Vision (**ACCVW**)

HONORS AND AWARDS

Travel Grant, IEEE/CVF International Conference on Computer Vision (ICCV), 2025

- Waived registration fees with partial scholarship from ICCV 2025, **\$1,100** (total)

18th Kwanjeong Educational Foundation Fellowship, KEF, 2023

- Winner, received **\$17,000** during 2 years of my graduate studies

Bang Sung-Yang Graduate Fellowship, POSTECH, 2023

- Admission with highest distinction, **\$1,200**

16th Kwanjeong Educational Foundation Fellowship, KEF, 2021

- Winner, received **\$16,500** during 2 years of my undergraduate studies

University Innovation Support Project Scholarship, UISPC, 2021

- Research fellowship, **\$750**

SW Specialization Scholarship, CAU, 2020 - 2022
- Research fellowship, **\$4,200** (total)

Da Vinci Scholarship, CAU, 2019
- Admission with highest distinction, **\$7,250**

Dean's List, CAU, 2019 - 2021
- 2019 Fall, 2020 Spring, 2020 Fall, 2021 Fall (4 times)

Awards

- **Best Paper Award (Grand Prize, 1st of 311 papers $\approx$ 0.3%),**
36th Workshop on Image Processing and Image Understanding (IPIU), 2024
- Excellence Prize in Research Report,
Innovation Center for Engineering Education, CAU, 2022
- Excellence Prize in Project,
Davinci Software Institute, CAU, 2021
- Best Paper Award,
Digital Contents Society, 2020
- Excellence Award in Project,
Innovation Center for Engineering Education, CAU, 2020
- Best Paper Award (top 10% papers),
International Next-generation Convergence Technology Association (INCA), 2020
- Best Paper Award (2nd Prize),
Humanities Research Institute, CAU, 2020

TEACHING EXPERIENCE

Pohang University of Science and Technology (POSTECH)

Teaching Assistant

Feb 2024 - Mar 2024

- CS2537/AIGS537 - Artificial Intelligence & Data Science (2024 Spring)

Ministry of National Defense, Republic of Korea

Instructor

Nov 2022 - Dec 2022

- Conducted lectures on AI projects for South Korean soldiers and short-term service officers
- Collaborated with *Elice, Inc.*

Samsung Electronics Co., Ltd

Instructor

Aug 2022 - Oct 2022

- Conducted lectures on data science and machine learning for newcomers (DS division)
- Collaborated with *Elice, Inc.*

LEADERSHIP

Graduate School of Dept. of CSE, POSTECH

President of Student Council

Feb 2024 - Feb 2025

- President of graduate school council of Department of Computer Science & Engineering, POSTECH.

Kwanjeong Educational Foundation Fellowship

Vice President

Jul 2023 - Jul 2024

- Vice president of 18th Kwanjeong Educational Foundation Fellowship members.

INVITED PRESENTATIONS

Shaping Visual Intelligence: The Past, Present, and Future of Computer Vision, Kangwon National University, Samcheok, Korea, Nov. 2025

LLMs as Research Assistants: A Practical Guide from My Latest Papers, VUNO R&D Seminar, Seoul, Korea, Jun. 2025

3D Geometric Shape Assembly via Efficient Point Cloud Matching, Samsung AI Forum (SAIF) 2024, Suwon, Korea, Nov. 2024

Combinative Matching for Shape Assembly, Samsung Global Technology Symposium (**GTS**) 2023, Seoul, Korea, Mar. 2023

LANGUAGE	Korean (native), English (fluent)
SKILLS	
REFEREES	Available on request.